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Marked Pulled/Immersed Ideal Interfaces and Actual Trace Welding

Research draft for the marked-width MLC program

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Abstract

We separate the coarse ideal marking on which a covering/immersion pair acts from a finer response/access trace overlay. For a finite pseudo-piece boundary atlas we give a necessary-and-sufficient compiler for branch-adapted ideal markings, including exact overlap and covering-refinement obstructions. We construct collared ideal-arc trace spaces, proper and end-proper end-germ trace objects, and the actual one-to-many welding obtained by immersion restriction, covering subdivision, and covering transport. The complete marked interface is functorial under words, normalized overlays, and certified final-root commuting squares. This closes a fixed-horizon direct-decoration interface module; universal horizon realization, H2, MBTT, UPTCR, and MLC remain open.

1 The marked correspondence

Let

$$F = (f, \iota) : V' \rightrightarrows V$$

be a holomorphic covering and a holomorphic immersion. For an ideal marking $\mathcal{I}$ of V, the KKL action condition is

$$\boxed{f^*\mathcal{I} \text{ refines } \iota^*\mathcal{I}.} \tag{1.1}$$

The immersion pullbacks may fail to exist or overlap, and properness or end-properness is additional data.

Definition 1.1 (Two-layer interface)**.** A finite marked interface is

$$\mathfrak{I}_F = (\mathcal{I}, \mathscr{O}, f^*\mathcal{I}, \iota^*\mathcal{I}, \mathfrak{p}, \mathfrak{e}, \mathcal{T}, \mathcal{W}, \mathfrak{F}),$$

where $\mathcal{I}$ is the action marking and $\mathscr{O}$ is a possibly finer trace/access/obstacle overlay. An endpoint of $\mathscr{O}$ lying inside an interval of $\mathcal{I}$ is response-only until it is successfully promoted by the action-marking compiler.

Warning 1.2. Trace visibility does not imply ideal-action visibility. No response subdivision is used as a KKL endpoint without proving (1.1).

2 Finite pulled/immersed marking compiler

Cut one ideal-boundary component at a retained phase endpoint. Let the target be tiled by micro-atoms $M_0, \dots, M_{N-1}$ and the source by $A_0, \dots, A_{m-1}$. Each source atom has a covering label $c(j)$; under the immersion it has either a proper target label $r(j)$ or the nonproper label $\perp$.

For a connected target atom union B, delete every proper source atom whose label lies outside B, and retain the remaining connected components that contain a proper atom labelled in B. Denote their union by $H_\iota(B)$.

Theorem 2.1 (Finite immersion pullback formula)**.** *For a boundary-monotone finite pseudo-piece atlas,*

$$\boxed{\iota^* B = H_\iota(B).} \tag{2.1}$$

Proof. A proper path subordinate to a retained component can land only on a proper atom mapped into B; intervening nonproper atoms have no target ideal landing interval. Hence its immersed image is subordinate to B. Conversely, every excluded atom is proper outside B, or is separated from all B-labelled atoms by such a blocker. A proper path landing at that blocker witnesses failure of subordination to B. Maximality in the definition of $\iota^* B$ proves the claim. □

A maximal nonproper source run with adjacent proper target labels p, q belongs to both pullback hulls unless p, q lie in one target interval.

Theorem 2.2 (Finest immersion-admissible marking)**.** *Join the proper labels adjacent to every maximal nonproper run and take the interval-convex transitive closure in the retained phase order. The resulting contiguous blocks form the unique finest target partition whose immersion pullbacks are pairwise disjoint. Every immersion-admissible marking is a coarsening of this partition.*

Proof. The overlap across every nonproper run forces the adjacent labels into one interval, and connectedness forces all intervening target micro-atoms into the same interval. After taking the closure, a nonproper run has proper neighbors in at most one target block. Proper atoms of other blocks are blockers in (2.1), so distinct pullbacks are disjoint. □

Let K be a connected component of an immersion pullback and let v_-, v_+ be its source boundary vertices. Write c_∂ for the covering map on source vertices.

Theorem 2.3 (Covering/immersion action criterion)**.** *For an immersion-admissible target marking $\mathcal{I}$, condition (1.1) holds if and only if*

$$\boxed{c_\partial(v_-), c_\partial(v_+) \in \partial\mathcal{I}} \tag{2.2}$$

for every connected component of every immersion pullback.

Proof. The covering marking is obtained by pulling back target endpoints. It refines the induced immersion marking exactly when every boundary of every immersion interval is one of these covering-preimage endpoints. □

Corollary 2.4 (Finite decision)**.** *For a finite mandatory endpoint set, enumerate the coarsenings of the finest immersion-admissible partition, retain the mandatory cuts, and test (2.2). This returns a finest valid action marking with all proper/end-proper flags, or an exact obstruction: a mandatory cut through an immersion-forced block, or a covering boundary image incompatible with every permitted coarsening.*

3 Collared ideal-arc trace spaces

Every pseudo-piece ideal tile arc lies on a universal-cover piece used in the source construction. For a proper finite ideal subarc J, choose a piecewise-analytic collar $C_Y(J)$ bounded by J, two crosscuts, and one interior cross-section.

Definition 3.1 (Collared trace)**.** Set

$$\mathcal{T}_Y(J) = \mathrm{Tr}_J\, H^1(C_Y(J))$$

with the minimum-energy quotient norm

$$\|\tau\|^2 = \inf_{\mathrm{Tr}_J\, u=\tau} \left(\int_{C_Y(J)} |\nabla u|^2 + \left| \int_{K_J} u \right|^2 \right), \tag{3.1}$$

where K_J is a fixed compact normalization set.

Theorem 3.2 (Ideal-arc trace theorem)**.** *$\mathcal{T}_Y(J)$ is a Hilbert trace space. In a conformal rectangle coordinate it is the usual $H^{1/2}$ trace space, and transporting the collar by a conformal isomorphism transports the Dirichlet seminorm isometrically.*

Proof. Apply the Sobolev trace theorem to the piecewise-analytic collar and use the minimum-energy quotient. A Riemann map identifies the collar with a rectangle; Dirichlet energy is conformally invariant in dimension two. The normalization term removes the constant kernel. □

For finitely many arcs, the compatible tuple space is the image of one H^1 function on the union of their collars and a finite connecting core. It is not replaced by an arbitrary direct sum.

If a component K of $\iota^* I$ is proper, retain its full compatible trace. If it is only end-proper, retain the compatible traces on its proper end intervals:

$$\mathcal{T}^{\mathrm{end}}_{V'}(K) = \mathcal{T}_{V'}(E_-(K), E_+(K)). \tag{3.2}$$

The nonproper middle remains in the interval bridge word and is not assigned a fictitious boundary trace.

4 Actual one-to-many welding

Let I be a target action interval. For each proper end arc $E \subset K \subset \iota^* I$, immersion restriction gives

$$R_{\iota,E} : \mathcal{T}_V(I) \longrightarrow \mathcal{T}_{V'}(E).$$

Because $f^*\mathcal{I}$ refines $\iota^*\mathcal{I}$, subdivide E into covering intervals E_ℓ. Degree-one covering transport gives

$$P_{f,E_\ell} : \mathcal{T}_{V'}(E_\ell) \longrightarrow \mathcal{T}_V(f(E_\ell)).$$

Definition 4.1 (Marked welding)**.**

$$\mathcal{W}_{F,I} := \bigoplus_{K,E,E_\ell} \big(P_{f,E_\ell} \circ \mathrm{Sub}_{E_\ell} \circ R_{\iota,E} \big). \tag{4.1}$$

The output index retains branch, component, interval, side, phase, endpoint flags, and the proper-or-nonproper bridge word.

Theorem 4.2 (Actual ideal welding)**.** *The operator (4.1) is continuous on the compatible trace spaces. It is the complete trace welding when F is proper on I, and the complete end-germ welding when F is end-proper on I. If $\nu(I)$ is the number of proper lifted end arcs, then*

$$\|\mathcal{W}_{F,I}\| \le \sqrt{\nu(I)} \tag{4.2}$$

in direct-sum Dirichlet norm.

Proof. Each factor is an actual trace restriction or conformal transport on a proper collar. The action condition prevents a covering endpoint from crossing an immersion interval. One parent trace is copied to at most $\nu(I)$ disjoint proper collars; conformal energy invariance gives (4.2). □

Theorem 4.3 (Word functoriality)**.** *For composable marked correspondences with matching intermediate action markings,*

$$\boxed{\mathcal{W}_{F_2\circ F_1,I} = \mathcal{W}_{F_2,F_{1*}I} \circ \mathcal{W}_{F_1,I},} \tag{4.3}$$

after canonical reassociation of the labelled compatible tuple spaces. The pulled intervals, properness flags, bridge words, endpoint flags, and cyclic orders compose in the same chronological order.

Proof. Both sides restrict the same finite-energy function along the same actual immersions, subdivide by the same covering endpoints, and transport through the same covering maps. Function restriction and map composition are associative. □

5 Repair and overlay rules

A trace-overlay endpoint not belonging to the action marking remains response-only. If a later KKL theorem requires it as an ideal endpoint, rerun the finite compiler.

Definition 5.1 (Slit-repair certificate)**.** A finite source/target analytic crosscut pair is a slit-repair certificate if it avoids all critical, protected, obstacle, port, and localization data; commutes with both maps; and separates the conflicting nonproper bridge after cutting.

Theorem 5.2 (Certified slit repair)**.** *Cutting along a slit-repair certificate lifts the covering and immersion to the cut surfaces, converts the certified split into a genuine ideal endpoint, and reduces the repaired interface to the finite compiler above.*

Proof. A finite analytic cut replaces each crosscut by two proper boundary sides. The commuting pullback condition lifts both maps to the slit surfaces. The former nonproper bridge is separated at the certified side, so the new endpoint is no longer inside an immersion-forced block. □

The theorem realizes a repair from a certificate; it does not construct such a certificate for every mixed block.

6 Normalized and final-root naturality

The binary pseudo-piece normalization transports the root collars, action markings, and overlay to every finite word. Source functoriality and (4.3) give exact normalized-overlay naturality; no physical-coordinate welding conjugacy is omitted.

Consider an old and enlarged pair with embeddings

$$A : V \to V^+, \qquad A' : V' \to V^{+'}$$

satisfying

$$\boxed{A \circ f = f^+ \circ A', \qquad A \circ \iota = \iota^+ \circ A'.} \tag{6.1}$$

Assume exact inverse image of all proof-used endpoint sets, transported collars, phase and side data, and injective cyclic-order-preserving ideal boundary maps.

Theorem 6.1 (Certified final-root naturality)**.** *Under (6.1), covering and immersion pullbacks commute with enlargement, the action condition is preserved, proper/end-proper flags are unchanged, and*

$$\boxed{A^\sharp_{\mathrm{out}} \mathcal{W}_{F^+,A(I)} = \mathcal{W}_{F,I} A^\sharp_{\mathrm{in}}} \tag{6.2}$$

after matching the labelled output tuple.

Proof. The first conclusions follow by taking inverse images in the two commuting squares. Trace restriction by A followed by old immersion pullback is the same function as enlarged immersion pullback followed by A'. The covering square gives the output identity. Subdivision and direct sums preserve these equalities, proving (6.2). □

A finite overlay refinement is natural under the same rule. Trace-only subdivision always commutes. Promotion to an action endpoint is allowed only when the compiler accepts it or a slit repair is supplied.

7 Fixed-horizon completion

Theorem 7.1 (Fixed-horizon marked binary interface)**.** *Fix one actual direct-decoration binary block and a finite source-authorized branch horizon. Freeze all mandatory ideal endpoints, trace/access/obstacle endpoints, proper/end-proper calls, and final-root generators. Assume every mandatory action endpoint passes the compiler or has a slit-repair certificate, and every enlargement generator supplies (6.1) with the stated boundary data. Then the complete horizon has:*
1. *actual covering-pulled and immersion-pulled ideal markings;*
2. *proof of the action condition on every branch;*
3. *complete proper/end-proper and endpoint-flag records;*
4. *valid collared ideal-arc and end-germ trace spaces;*
5. *actual one-to-many welding and bridge words;*
6. *exact word, normalization, overlay, and final-root naturality;*
7. *a finite first-failure certificate otherwise.*

Proof. Run the finite marking compiler at every branch, apply certified repairs, build the collared trace spaces, and use the welding and naturality theorems above. The horizon is finite, so the union of all records is finite. □

8 Audit and promotion boundary

The companion compiler was checked against an independent exhaustive search of every target endpoint subset in 1,500 randomized systems with at most seven target micro-atoms. It compiled 1,221 systems, obstructed 279, and had zero false positives, false negatives, or maximal-marking mismatches. A second exact-integer audit checked 1,000 welding composition and enlargement squares with zero failures.

The theorem closes a fixed-horizon marked-interface module. It does not build the complete mixed Wave horizon, universally realize every mandatory endpoint or slit, bound lift multiplicity, construct a universal response/WAD/Value quotient, prove MBTT, or prove MLC.

9 Next obligation

The next source theorem is

`UNIVERSAL_MANDATORY_ENDPOINT_VISIBILITY_AND_REPAIR.`

It must prove that every proof-required ideal endpoint in every mixed block is compiler-visible or has a collision-free slit repair, remains natural under final-root enlargement, has universally bounded proper-lift multiplicity, and factors with all response/WAD/Value outputs through one uniformly finite future quotient.

References

[1] J. Kahn and M. Lyubich, *A priori bounds for some infinitely renormalizable quadratics. II. Decorations*, arXiv:math/0609046.

[2] J. Kahn, A. Kapiamba, and M. Lyubich, *MLC for parabolically bounded primitive renormalization*, arXiv:2606.27272.